Week 2: Control Flow, Functions & First Dataset Exploration

Welcome to Week 2.

Last week, you built your first Python program using variables and basic operations. This week, you'll learn how to make decisions in your code (if/else statements), repeat actions (loops), and organise code into reusable functions.

More importantly, this is where the supply chain dataset enters. You'll load real data for the first time and start exploring it. You'll work with this dataset for the next 11 weeks.

Why this matters: Control flow and functions are the backbone of all programming. You need them to process data and make decisions based on it. Exploring the dataset shows you how real data looks and feels, including its problems.

Project Goal

Write a Python notebook that uses if/else statements, loops and functions to process order and supplier data. Then load the supply chain dataset and explore its structure.

Part A: What You Will Learn

Essential Concept 1: If/Elif/Else Statements

What they are: Decision-making in code.

Why you need them: To make your program respond differently to different situations, for example "if the order is above 100,000, it needs a manager's approval".

Essential Concept 2: Comparison and Logical Operators

What they are: Tools to compare values (`==`, `!=`, `<`, `>`, `<=`, `>=`) and combine conditions (`and`, `or`, `not`).

Why you need them: To build the conditions inside your if/else statements.

Essential Concept 3: For Loops

What they are: Repeat an action for each item in a collection.

Why you need them: To process 3,000 orders without writing the same code 3,000 times.

Essential Concept 4: Functions

What they are: Reusable blocks of code that do one job. A function has a name, parameters (inputs) and a return value (output).

Why you need them: To avoid repeating code and keep your notebook organised.

Essential Concept 5: Reading CSV Files

What they are: CSV means Comma-Separated Values: a simple text file where each line is a row and commas separate the columns. Python's built-in `csv` module reads them.

Why you need it: The supply chain dataset is a CSV file.

Essential Concept 6: Converting Text to Numbers

What it is: Everything read from a CSV arrives as text. Use `float("6062.06")` or `int("94")` to turn it into a number before doing maths or comparing sizes.

Part B: Video Resources

Video	Topic	Link	Focus
1	Conditionals and Booleans, Corey Schafer (16 min)	Watch	if/elif/else, comparison and logical operators
2	If Statements, Bro Code (8 min, optional)	Watch	A shorter recap of if/else
3	For Loops in Python, Alex The Analyst (9 min)	Watch	Looping through lists
4	Functions in Python, Bro Code (11 min)	Watch	Parameters and return values
5	CSV Module, Corey Schafer (16 min)	Watch	Reading CSV files with `csv.DictReader`

Meet the Dataset

Download `supply_chain_dataset.csv` from the course folder and put it in the same folder as your notebook (in Colab, upload it with the folder icon on the left). It contains purchase orders that a manufacturer placed with five suppliers in 2022 and 2023.

Column	Meaning
order_id	Order number, e.g. ORD-0001
order_date	Date the order was placed
product_category	What was ordered (Motors, Pumps, Sensors, ...)
quantity	Units ordered
unit_price	Price per unit

Column	Meaning
total_cost	Amount spent on the order (should equal quantity x unit_price)
supplier_name	The supplier (there are five: SupplierA to SupplierE)
supplier_location	Supplier's country
lead_time_days	Delivery time the supplier promised, in days
warehouse_destination	Warehouse receiving the order
delivery_date	Date the order arrived (blank if not delivered yet)
actual_delivery_days	Days the delivery actually took
delivery_status	Delivered, Delayed (arrived late), Cancelled, In-Transit or Pending
on_time	Yes if delivered within the promised lead time, otherwise No
product_defects	Number of defective units received
return_status	No Return, Partial Return or Full Return

> This data is deliberately messy, like real company data. Don't fix anything yet: this week you're only looking. You'll clean it in Weeks 4 and 7.

Part C: The Tasks

Task 1: Order Approval Logic

Write a function called `approval_message(order_value)` that returns:

- "Needs manager approval" if the value is above 100,000
- "Needs supervisor approval" if it is above 50,000
- "Auto-approved" otherwise

Test it with an order value of 72,500.

Task 2: Process Multiple Orders

Create the list `order_values = [12000, 55000, 150000, 49999, 100001, 8000]`. Use a for loop to print each value with its approval message.

Task 3: Supplier Status Function

Create a function `check_supplier_status(supplier_name, on_time_rate)` that returns the supplier's name with a status:

- "Reliable" if the on-time rate is 0.90 or higher

- "Monitor" if it is 0.80 or higher
- "At Risk" otherwise

Test it with at least 3 suppliers, for example `("SupplierA", 0.94)`, `("SupplierB", 0.86)` and `("SupplierE", 0.74)`.

Task 4: Load the Supply Chain Dataset

Use this code to load the file into a list of rows:

```
import csv

with open("supply_chain_dataset.csv", newline="") as file:
    reader = csv.DictReader(file)
    rows = list(reader)
```

Each row is a dictionary, so `rows[0]["total_cost"]` gives the first order's cost. Write code to print:

- The column names (hint: `rows[0].keys()`)
- The number of data rows
- The first 5 rows

Task 5: Explore the Dataset

Using loops and if statements, answer these questions and print each answer with a clear label:

1. How many rows of data are there?
2. How many different supplier names are written in the file? Print them. (The company says it has five suppliers. What do you notice?)
3. What is the highest `total_cost`? (Remember `float()`.)
4. What is the lowest `total_cost`? Does it make sense?
5. How many orders have the `delivery_status` "Delivered"?

Part C Continued: Extra Challenges

Challenge 1: Write a function that takes an order value and a supplier's on-time rate, and returns a different approval level based on both. For example, reject orders above 100,000 from suppliers below 0.80.

Challenge 2: Find all orders from SupplierA, however the name is written. Hint: `name.lower().replace(" ", "")` turns "Supplier A" into "suppliera". Count them and add up their total cost.

Challenge 3: Loop through the dataset and find rows with empty values (`row[column] == ""`). Which columns have gaps, and how many?

What You've Learned This Week

- If/elif/else statements and conditional logic
- Comparison and logical operators
- For loops
- Functions: definition, parameters, return values
- Reading CSV files with `csv.DictReader`
- Converting text to numbers
- Answering questions about real data with code

Common Mistakes

Mistake 1: Forgetting the colon after if, for or def

```
# WRONG
if age > 65
    print("Retired")

# RIGHT
if age > 65:
    print("Retired")
```

Mistake 2: Wrong indentation

```
# WRONG
if score > 50:
print("Passed")

# RIGHT
if score > 50:
    print("Passed")
```

Mistake 3: Using = instead of == to compare

```
# WRONG
if name = "Alex":

# RIGHT
if name == "Alex":
```

Mistake 4: Comparing text instead of numbers

```
# WRONG: "9000" > "10000" is True, because text is compared letter by letter
if row["total_cost"] > "10000":

# RIGHT
if float(row["total_cost"]) > 10000:
```

Mistake 5: Forgetting to return (or call) your function

```
# WRONG: calculates but gives nothing back
def calculate_vat(amount):
    vat = amount * 0.075

# RIGHT
def calculate_vat(amount):
    vat = amount * 0.075
    return vat

print(calculate_vat(1000))
```

Part D: Deliverables for Week 2

1. Your notebook with all 5 tasks, clean comments, and no errors, uploaded to your GitHub repository.
2. A Dataset Summary (one paragraph, in a text cell):
    - How many rows and columns are in the data?
    - What problems did you notice (supplier names, the lowest cost, empty values)?
    - Three questions you would like to answer with this data.

What to post: Screenshots of your code and output, including the dataset loaded and explored.

Part E: Submit Your Work

Submission Period: Friday 7am to Saturday 5pm (WAT)

Where to Submit: X (Twitter) or LinkedIn (or both!), plus the submission form that the Group Admin will drop in the WhatsApp group. Paste the link to your GitHub repository in the form.

For X (Twitter) submissions, please tag:

1. @TDataImmersed (TDI Official page)
2. @DabereNnamani (Annie)
3. @ThePSF
4. Use the hashtag #TheDataImmersed

For LinkedIn submissions, please tag:

1. TDI page
2. Annie Nnamani

Next week: Data structures (lists, dictionaries and sets) and mapping the supplier network!